



United International University (UIU)

Dept. of Computer Science and Engineering (CSE)

Final Exam

Year: 2025

Semester: Spring

Course Code: CSE 2217

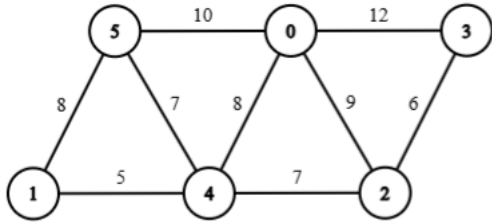
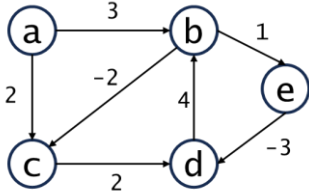
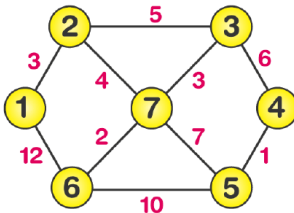
Title: Data Structure and Algorithms II

Marks: 40

Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. All questions are of values indicated on the right-hand margin.

1.	a. Apply Kruskal's Algorithm to find the Maximum Spanning Tree (MST) of the following graph and calculate its total weight.  b. If an undirected graph contains 50 nodes and 1000 edges, which algorithm would you choose between Prim's and Kruskal's to find the minimum spanning tree (MST) of that graph? Justify your answer by mentioning the time complexities of the algorithms. c. “Minimum spanning tree of an undirected graph is unique” - Is this statement true or false? Justify your answer by constructing a graph with 4 nodes.	[4] [2] [1]
2.	a. Find the shortest path from ‘a’ to all other vertices in the following directed graph using the appropriate algorithm.  b. If a graph contains a negative weight cycle, why can't we find a valid shortest path from the source to some vertices? Explain with an example. c. Which algorithm you would prefer among BFS, Dijkstra and Bellman-Ford to find the shortest path in the following graph. Justify your answer with mathematical reasoning. 	[5] [3] [2]
3.	a. How does path compression heuristic improve the time complexity of find operation in disjoint set data structure? Explain with a proper example.	[2]

	b. Table 1 shows the parent array of a Disjoint set (Rooted tree implementation). Perform the following operations sequentially using path compression and union-by-rank heuristic: i. Draw the disjoint set forest. ii. What will be returned by Find-Set(4), and Find-Set(6)? Show details. iii. Redraw the forest after Union(6, 9). iv. Redraw the forest after Union(9, 5). <table border="1"><tr><td>Index</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>Parent</td><td>0</td><td>0</td><td>1</td><td>0</td><td>3</td><td>4</td><td>7</td><td>8</td><td>8</td><td>9</td><td>4</td><td>9</td></tr></table> Table 1: Disjoint set (Rooted tree implementation)	Index	0	1	2	3	4	5	6	7	8	9	10	11	Parent	0	0	1	0	3	4	7	8	8	9	4	9	[2] [2] [1] [1]
Index	0	1	2	3	4	5	6	7	8	9	10	11																
Parent	0	0	1	0	3	4	7	8	8	9	4	9																
4.	a. When would you choose a Direct Address Table over a Hash Table using Chaining? Explain with an example. b. Consider the hash function: $h(k, i) = (h'(k) + i^2) \bmod 13$ where, $h'(k) = (2k + 1) \bmod 13$ By proper calculation, redraw Table 2 and illustrate the following sequence of operations, including accurate hash value calculations. i. Insert 38 ii. Insert 51 iii. Delete 19 and replace with NIL iv. Search 51 v. If your search of 51 fails despite being present at the table, what might be the possible reason? Explain how you can modify the delete operation to prevent this from happening. Currently, the table contains the following: <table border="1"><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>19</td><td>32</td><td>NIL</td><td>NIL</td><td>21</td><td>NIL</td><td>NIL</td><td>42</td><td>NIL</td><td>NIL</td><td>NIL</td><td>70</td><td>NIL</td></tr></table> Table 2: A hash table	0	1	2	3	4	5	6	7	8	9	10	11	12	19	32	NIL	NIL	21	NIL	NIL	42	NIL	NIL	NIL	70	NIL	[2] [1] [1] [1] [1] [2]
0	1	2	3	4	5	6	7	8	9	10	11	12																
19	32	NIL	NIL	21	NIL	NIL	42	NIL	NIL	NIL	70	NIL																
5.	a. What is a spurious hit in the Rabin-Karp algorithm? Explain with an example. b. You are working on a search feature for an e-book reader application. One of the users wants to find whether a specific word segment exists in a book's paragraph. The paragraph contains the text: "sunbeam". The user wants to search for the pattern: "eam". You need to efficiently identify all the occurrences of the pattern within the text. The hash code for each character is given as follows: <table border="1"><tr><td>character</td><td>S</td><td>U</td><td>N</td><td>B</td><td>E</td><td>A</td><td>M</td></tr><tr><td>code</td><td>9</td><td>5</td><td>3</td><td>2</td><td>7</td><td>1</td><td>4</td></tr></table> The hash function is defined as: $hash(s) = (s[m - 1] * d^{m-1} + s[m - 2] * d^{m-2} + \dots + s[1] * d^1 + s[0] * d^0) \bmod q$ where, m = length of the string, d = 7, q = 29 Show the indices of both valid hits and spurious hits (if any) using the aforementioned hash function with detailed simulation.	character	S	U	N	B	E	A	M	code	9	5	3	2	7	1	4	[2] [5]										
character	S	U	N	B	E	A	M																					
code	9	5	3	2	7	1	4																					



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Final Exam Total Marks: 40 Fall 2024

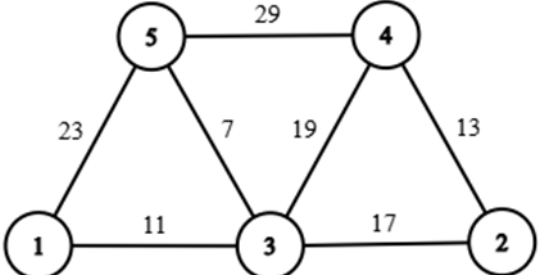
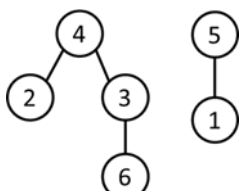
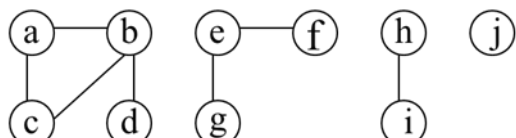
Course Code: CSE 2217

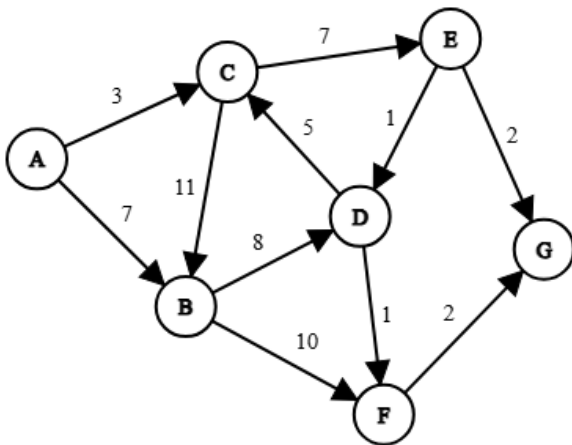
Course Title: Data Structure and Algorithms II

Time: 2 hours

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There are **five** questions. **Answer all of them.** Show full simulation/tabulations wherever necessary. Figures in the right-hand margin indicate full marks.

1.	(a) Apply Prim's Algorithm to find the Maximum Spanning Tree (MST) on the following graph and calculate its total weight.  (b) Given an undirected graph with 50 vertices and 200 edges, which algorithm would you choose between Prim's and Kruskal's algorithms to find the MST? Justify your answer with proper mathematical reasoning. (c) Given a connected undirected graph with 6 vertices and 10 edges, what is the maximum number of edges that can be included in a Minimum Spanning Tree (MST)?	[4] [2] [1]
2.	(a) Draw the resultant forest after calling UNION(5, 6) and after that draw the resultant forest again after calling FIND-SET(1) on the disjoint-sets of the following figure. You must use the <i>union-by-rank</i> and the <i>path-compression</i> heuristics.  (b) Using disjoint-set data structures, determine whether node c and node f are in the same connected component.  (c) Suppose you are asked to utilize a disjoint set data structure to implement Kruskal's algorithm to find MST from a given graph. Explain, with an example, how disjoint set data	[3] [2] [3]

	structure would help you to perform the following operations: i) joining two trees in the forest ii) detecting a cycle													
3	(a) If we increase the weights of every edge in a graph by the same amount does the shortest path change? Justify your answer by showing an example. (b) Apply Dijkstra's algorithm to find the shortest path from source node A to every node in the following directed graph.  (c) "Every connected directed acyclic graph (DAG) has exactly one topological ordering" - true or false? Design a graph G with exactly 4 vertices that justifies your reasoning. (d) The Bellman-Ford algorithm tries to relax every edge in each iteration. If there are n vertices, explain why the algorithm requires (n-1) iterations to correctly calculate the lengths of all shortest paths in the graph.	[2] [4] [1+2] [2]												
4	(a) What is a spurious hit in the Rabin-Karp algorithm? How does the occurrence of spurious hits impact the time complexity of the Rabin-Karp algorithm? (b) You are tasked with building a plagiarism detection system to check if short patterns from a reference document are present in a student's submission. Suppose the reference document contains the text "academy" and you want to verify if the pattern "emy" appears in it. You need to apply Rabin-Karp's algorithm (using the rolling hash technique) to find whether the pattern exists in the text or not. The hash code for each character is given as follows: <table><tr><td>a</td><td>c</td><td>d</td><td>e</td><td>m</td><td>y</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr></table> The hash function is defined as: $h(xyz) = (x + y + z) \% 7$ Where, x, y and z represent the hash codes for each character. You must calculate the hash values for the pattern and for all substrings in the text "academy" using rolling hash technique. Find the valid hits and spurious hits(if any).	a	c	d	e	m	y	1	2	3	4	5	6	[2] [4]
a	c	d	e	m	y									
1	2	3	4	5	6									

5	(a) In Chaining, what is the average cost of a successful search? (the load factor is defined as α) (b) Given that memory is not an issue to be concerned about, what is the best possible hashing scheme and why? (c) Consider an open-addressing hash table as shown below. The table already contains some data items and other empty slots. Assume that collisions are handled by Quadratic probing using the hash function: $h(k, i) = (h'(k) + 2i^2) \bmod 13$, where $h'(k) = (k + 7) \bmod 13$ By showing detailed calculations, redraw the table after (i) Insert 47 (ii) Insert 64 (iii) Delete 12 (replace with NIL) (iv) Search 38. <table border="1"><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td></td><td>44</td><td></td><td></td><td></td><td></td><td>12</td><td></td><td>38</td><td></td><td></td><td></td><td>70</td></tr></table> Although 38 had been present at the hash table, your search would fail here. Explain how one can modify the operations such that this will be prevented.	0	1	2	3	4	5	6	7	8	9	10	11	12		44					12		38				70	[1] [2] [1] [1] [1] [1]
0	1	2	3	4	5	6	7	8	9	10	11	12																
	44					12		38				70																